

Agile Planning Logic & Estimation

1. Agile Framework Core Definitions

- Sprint:** A fixed period or duration in which a team works to complete a set of tasks.
- Epic:** A big task or project that is too large to complete in one sprint. It is broken down into smaller tasks (stories) that can be completed over multiple sprints.
- Story / User Story:** A small task that forms a constituent part of an Epic.
- Story Point:** A numerical estimation representing how much effort a story takes to complete, typically structured around the Fibonacci sequence (1, 2, 3, 5) where tasks scale from Very Easy to Difficult.

2. Sprint Distribution and Backlog Scope

Sprint 1

Epic Block	User Story Task	Story Points
Epic 1: Data Collection	Gathering Historical Weather Data (USN1)	2
Epic 1: Data Collection	Loading Dataset into Python (USN2)	1
Epic 2: Data Preparation	Handling Missing Values (USN3)	3
Epic 2: Data Preparation	Feature Selection & Engineering (USN4)	3
Epic 2: Data Preparation	Data Cleaning and Encoding (USN5)	3
Total Story Points (Sprint 1)		12

Sprint 2

Epic Block	User Story Task	Story Points
Epic 3: Model Development	Train Decision Tree Model (USN6)	2
Epic 3: Model Development	Train Random Forest Model (USN7)	2
Epic 3: Model Development	Train KNN Model (USN8)	2
Epic 3: Model Development	Train XGBoost Model & Compare Accuracy (USN9)	4
Epic 4: Web Application Dev	Develop Flask Web Application (USN10)	5
Epic 5: Testing & Deployment	Model Testing and IBM Cloud Deployment (USN11)	5
Total Story Points (Sprint 2)		20

3. Team Velocity Calculation

Velocity Metrics Summary

- Total Accumulated Story Points = 32 points (Sprint 1: 12 pts + Sprint 2: 20 pts)
- Total Planned Sprints = 2 Sprints

Formula: Velocity = Total Story Points / Number of Sprints

Calculation: $32 / 2 = 16$ Story Points per Sprint